

Advanced Communication Networks

شبکه‌های ارتباطی پیشرفته

Course Syllabus

Objectives, Modules, Schedule and Grading

طرح درس



Lecture Notes

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Course Information

Field	Details
Course Title	Advanced Communication Networks
Credits	3
Total Contact Hours	48 hours
Duration	16 weeks × 3 hours/week
Format	Lectures + Labs + Project
Target Students	Senior undergraduate / graduate EE, Telecom Engineering

Course Objectives

By the end of this course, students will be able to:

1. Trace the complete evolution of mobile networks from GSM (2G) through UMTS (3G), LTE (4G), 5G NR/SA, to emerging 6G concepts.
2. Explain WHY each generation was developed, WHAT problem it solved, and HOW traffic flows through each architecture.
3. Analyze core network architectures: circuit-switched, packet-switched, all-IP EPC, and service-based 5G Core.
4. Describe mobility management, handover mechanisms, and their evolution across generations.
5. Understand network slicing, QoS frameworks, SDN, and NFV in modern networks.
6. Explain 5G Core procedures (registration, authentication, PDU session) at the signaling level.
7. Critically assess 6G research directions (ISAC, RIS, AI-native, cell-free MIMO, sub-THz).

Module List (26 Modules)

#	Module Title	File
1	GSM, GPRS, and EDGE	01_gsm_gprs_edge.md
2	2G Circuit Switching	02_circuit_switching.md
3	2G/3G Core Networks	03_2g_3g_core.md
4	Mobility and Handover	04_mobility_handover.md
5	4G LTE Architecture	05_4g_lte.md
6	E-UTRAN (LTE Radio Access)	06_eutran.md
7	LTE Scheduling	07_lte_scheduling.md
8	Random Access Channel (RACH)	08_random_access.md
9	Evolved Packet Core (EPC)	09_epc.md
10	Diameter Protocol	10_diameter.md
11	Home Subscriber Server (HSS)	11_hss.md
12	Mobility Management Entity (MME)	12_mme.md
13	S-GW and P-GW	13_sgw_pgw.md
14	5G Networks Introduction	14_5g_networks.md
15	Software Defined Networking (SDN)	15_sdn.md
16	Network Function Virtualization (NFV)	16_nfv.md
17	3GPP Standardization	17_3gpp.md
18	5G Service-Based Architecture (SBA)	18_sba.md
19	5G Core Architecture	19_5g_core.md
20	5G Procedures	20_5g_procedures.md
21	Network Slicing	21_network_slicing.md
22	Quality of Service (QoS)	22_qos.md
23	User Plane	23_user_plane.md
24	Control Plane	24_control_plane.md
25	Network Evolution — Grand Synthesis	25_network_evolution.md
26	Future 6G Technologies	26_future_6g.md

Supporting files: [00_prerequisites.md](#) (background review), [00_syllabus.md](#) (this file)

Grading Breakdown

Component	Weight	Details
Exams	50%	Midterm (20%) + Final (30%)
Labs	30%	6 lab assignments (5% each)
Project	20%	Proposal (5%) + Report & Demo (15%)

Recommended Textbooks

1. **Sesia, S., Tufik, I., & Baker, M.** — *LTE – The UMTS Long Term Evolution: From Theory to Practice*, 2nd Ed., Wiley.
2. **Dahlman, E., Parkvall, S., & Sköld, J.** — *5G NR: The Next Generation Wireless Access Technology*, Academic Press.
3. **Holma, H. & Toskala, A.** — *LTE for UMTS*, Wiley.
4. **3GPP Specifications** — TS 23.501 (5GS), TS 23.502 (procedures), TS 24.501 (NAS), TS 38.300 (NR) — at 3gpp.org.
5. **Rappaport, T. S.** — *Wireless Communications: Principles and Practice*, 2nd Ed., Prentice Hall.

Week-by-Week Schedule

Week	Modules	Topic Focus
1	1–2	GSM architecture, GPRS/EDGE, circuit switching fundamentals
2	3	2G/3G core networks, UMTS architecture, CS/PS domains
3	4	Mobility management, handover types (2G→5G evolution)
4	5–6	LTE architecture, E-UTRAN, protocol stack; Lab 1 assigned
5	7–8	LTE scheduling, RACH procedure
6	9	Evolved Packet Core (deep dive); Lab 2 assigned
7	10–11	Diameter protocol, HSS, authentication
8	12–13	MME procedures, S-GW/P-GW, GTP; Midterm Review
9	—	MIDTERM EXAM
10	14	5G introduction, NR, NSA vs SA; Lab 3 assigned
11	15–16	SDN, NFV, cloud-native networks
12	17–18	3GPP standardization, Service-Based Architecture; Lab 4 assigned
13	19–20	5G Core architecture, 5G procedures (registration, PDU session)
14	21–22	Network slicing, QoS evolution (QCI→5QI); Lab 5 assigned
15	23–25	User plane, control plane, network evolution synthesis; Lab 6 assigned
16	26	Future 6G technologies, project presentations & final review
17	—	FINAL EXAM

Lab Assignments

Lab	Week	Topic	Related Modules
1	4	LTE Link Budget & Path Loss	5, 6
2	6	EPC Signaling (Attach/Detach trace)	9, 12
3	10	5G NR Scheduling & Resource Allocation	7, 14
4	12	5G Core SBA Service Discovery	18, 19
5	14	Network Slicing Configuration	21, 22
6	15	Traffic Flow Tracing (User Plane)	23, 24

Project

Students work individually or in pairs (max 2) on a semester project. Topics may include:

- Comparative analysis of LTE vs 5G for a specific use case
- Network slicing design for an enterprise scenario
- Implementation of a scheduling algorithm (using the wireless_lab simulator)
- 6G research topic survey and feasibility analysis
- Signaling analysis using real-world packet captures

Deliverables: Proposal (Week 13), Final Report + Demo/Presentation (Week 16)

Academic Integrity

All submitted work must be original. Collaboration on labs is encouraged for discussion but each student submits individually. Plagiarism will result in a zero for the assignment and referral to the academic integrity board.

Last updated: August 2026